INTERNATIONAL GCSE MATHEMATICS A · MODULAR · UNIT 1 HIGHER

Expected Paper ELT/U1X Mark Scheme

Step by step, with the marks



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M = method mark, A = accuracy mark (needs the method), B = mark for a correct value on its own. Questions 19–26 also show THE IDEA: the one thought that unlocks them.

ANSWERS AT A GLANCE

1	shown: $\frac{62}{15} \times \frac{10}{31} = \frac{4}{3}$
2	20
3	{2, 4, 6, 8, 12, 24}; e.g. {2, 4, 6}
4	$4x^{11}y^{-6}$
5	9
6	$x = -\frac{11}{2}$
7	1000 N/m^2
8	$4x^3 - 4x^2 - 15x + 18$
9	8.91 cm
10	(b) 0.16
11	shown
12	$x = 1$, $x \approx 2.7$
13	$20 + 4\pi$ cm
14	$k = 30$
15	13.2 cm
16	$(4 + \sqrt{5})$ cm
17	$\frac{2x + 1}{x}$
18	$(2x - 5)(3x + 4)$; $\frac{5}{2}$, $-\frac{4}{3}$
19	$2(x + 2)^2 + 3$; max $\frac{10}{3}$ at $x = -2$
20	$5 + 2\sqrt{3}$
21	$k = 60$; $\frac{28}{85}$
22	$3x + 4y = 36$; $D(8, 3)$
23	$r = 7.87$
24	$n = 10$
25	22.4°
26	$12\sqrt{3} + 2\pi\sqrt{3}$ cm

1 Fractions without a calculator

3 marks

1 **IMPROPER FRACTIONS, ADD**

$\frac{7}{3} + \frac{9}{5} = \frac{35}{15} + \frac{27}{15} = \frac{62}{15}$ **M1**

2 **DIVIDE**

$\frac{62}{15} \div \frac{31}{10} = \frac{62}{15} \times \frac{10}{31}$ **M1**

3 **SIMPLIFY**

$= \frac{620}{465} = \frac{4}{3} = 1\frac{1}{3}$ **A1**

2 Estimation

2 marks

4 **ROUND**

$\frac{\sqrt{50 \times 2}}{0.5} = \frac{10}{0.5}$ **M1**

ANSWER 20 **A1**

3 Sets

3 marks

5 **LIST**

$A = \{1, 2, 3, 4, 6, 8, 12, 24\}$; $A \cap B'$ = the even factors **B1**

6 *D*

Even factors of 24 (so none odd), must contain 4, and no other square (so not 16 — not in A anyway). Choose 2 from $\{2, 6, 8, 12, 24\}$. **B2**

ANSWER (a) $\{2, 4, 6, 8, 12, 24\}$ (b) e.g. $\{2, 4, 6\}$

4 Index laws

2 marks

7 **POWER OF A POWER**

$(2x^3y^{-1})^4 = 16x^{12}y^{-4}$ **M1**

ANSWER $\frac{16x^{12}y^{-4}}{4xy^2} = 4x^{11}y^{-6}$ **A1**

5 Probability — adding counters

3 marks

8 **LET r RED BE ADDED, SO $3r$ BLUE**

total = $20 + 4r$ **M1**

9 **SET UP**

$\dfrac{12}{20+4r} = \dfrac{3}{8} \Rightarrow 96 = 60 + 12r \Rightarrow r = 3$ **M1**

ANSWER $3 \times 3 = 9$ blue counters **A1**

6 Index equation

3 marks

10 **BASE 3**

$3^{-2x} \times 3^{3x+6} = 3^{1/2}$ **M1**

11 **EQUATE**

$x + 6 = \frac{1}{2}$ **M1**

ANSWER $x = -\frac{11}{2}$ **A1**

7 Pressure — turning the block

3 marks

12 **THE FORCE DOES NOT CHANGE**

$F = 400 \times (2 \times 1.5) = 1200$ N **M1**

13 **SMALLEST FACE**

$1.5 \times 0.8 = 1.2$ m^2 **M1**

ANSWER $1200 \div 1.2 = 1000$ N/m^2 **A1**

8 Expanding

3 marks

14 **SQUARE FIRST**

$(2x - 3)^2 = 4x^2 - 12x + 9$ **M1**

15 **TIMES** $(x + 2)$

$4x^3 + 8x^2 - 12x^2 - 24x + 9x + 18$ **M1**

ANSWER $4x^3 - 4x^2 - 15x + 18$ **A1**

9 Two right-angled triangles

4 marks

16 **TRIANGLE ABC: AB IS ADJACENT TO 34°**

$AC = \dfrac{8.5}{\cos 34^\circ} = 10.25\ldots$ **M1 A1**

17 **TRIANGLE ACD: AC IS ADJACENT TO 41°**

$CD = 10.25\ldots \times \tan 41^\circ$ **M1**

ANSWER 8.91 cm **A1**

10 Tree diagram

4 marks

18 **(A)**

bus 0.4; walks: not late 0.9; bus: late 0.25, not late 0.75 **B2**

19 **(B) TWO LATE PATHS**

$0.6 \times 0.1 + 0.4 \times 0.25$ **M1**

ANSWER 0.16 **A1**

11 Recurring decimal

2 marks

20 **LINE UP THE TAILS**

$x = 1.23636\ldots$, $10x = 12.3636\ldots$, $1000x = 1236.3636\ldots$ **M1**

21 **SUBTRACT**

$990x = 1224 \Rightarrow x = \dfrac{1224}{990} = \dfrac{68}{55}$ **A1**

12 Graph and a straight line

7 marks

22 **(A)**

$x = 1$: 3; $x = 2$: 5; $x = 3$: 9.67 **B2**

23 **(B)**

Plot all points; smooth curve with its lowest point near $(1, 3)$. **B2**

24 **(C) DIVIDE BY x TO MATCH THE GRAPH**

$x^3 - 3x^2 + 2 = 0 \Rightarrow x^2 - 3x + \dfrac{2}{x} = 0 \Rightarrow x^2 + \dfrac{2}{x} = 3x$ **M1**

25 **DRAW $y = 3x$**

through $(0, 0)$ and $(3, 9)$ **M1**

ANSWER $x = 1$ and $x \approx 2.7$ (exact: $1 + \sqrt{3} = 2.73$) **A1**

13 Sector backwards

4 marks

26 **AREA GIVES r**

$\frac{72}{360}\pi r^2 = 20\pi \Rightarrow r^2 = 100,\ r = 10$ **M1 A1**

27 **ARC**

$\frac{72}{360} \times 2\pi \times 10 = 4\pi$ **M1**

ANSWER $20 + 4\pi$ cm **A1**

14 Bounds with a subtraction

4 marks

28 **UPPER BOUND: BIG TOP, SMALL BOTTOM**

$\frac{64.5 - 23.55}{1.35} = 30.33\ldots$ **M1**

29 **LOWER BOUND**

$\frac{63.5 - 23.65}{1.45} = 27.48\ldots$ **M1**

30 **COMPARE**

To 2 s.f.: 30 and 27 — different. To 1 s.f.: 30 and 30. **A1**

ANSWER $k = 30$, because both bounds round to 30 to 1 s.f. **A1**

15 Quadrilateral — cosine rule then sine rule

5 marks

31 **COSINE RULE IN ABC**

$AC^2 = 7^2 + 10^2 - 2(7)(10)\cos 110^\circ = 196.88\ldots,\ AC = 14.03\ldots$ **M1 A1**

32 **THIRD ANGLE OF ACD**

$D = 180 - 48 - 62 = 70^\circ$ **B1**

33 **SINE RULE**

$\frac{AD}{\sin 62^\circ} = \frac{14.03\ldots}{\sin 70^\circ}$ **M1**

ANSWER 13.2 cm **A1**

16 Surds — width of a rectangle

3 marks

34 **WIDTH = AREA ÷ LENGTH; RATIONALISE**

$$\frac{(17+7\sqrt{5})(3-\sqrt{5})}{(3+\sqrt{5})(3-\sqrt{5})}$$ **M1**

35 **EXPAND**

$$\frac{51-17\sqrt{5}+21\sqrt{5}-35}{9-5}=\frac{16+4\sqrt{5}}{4}$$ **M1**

ANSWER $(4+\sqrt{5})$ cm **A1**

17 Algebraic fractions — division

4 marks

36 **FACTORISE ALL FOUR**

$(x-3)(x+3)$, $(2x-1)(x+3)$, $x(x-3)$, $(2x-1)(2x+1)$ **M1 M1**

37 **FLIP AND CANCEL**

$$\frac{(x-3)(x+3)}{(2x-1)(x+3)}\times\frac{(2x-1)(2x+1)}{x(x-3)}$$ **M1**

ANSWER $\dfrac{2x+1}{x}$ **A1**

18 Factorise and solve

3 marks

ANSWER (a) $(2x-5)(3x+4)$ **M1 A1** (b) $x=\frac{5}{2}$ or $x=-\frac{4}{3}$ **B1**

19 Completing the square — a maximum

4 marks

THE IDEA a fraction is biggest when its bottom is smallest.

38 **(A)**

$2(x^2+4x)+11=2\left[(x+2)^2-4\right]+11=2(x+2)^2+3$ **M1 A1**

39 **(B) SMALLEST BOTTOM**

$2(x+2)^2+3\geq 3$, smallest value 3 when $x=-2$ **M1**

ANSWER maximum $\dfrac{10}{3}$ at $x=-2$ **A1**

20 Equation with surds

4 marks

THE IDEA collect the x terms, factorise x out, then rationalise.

40 **COLLECT**

$x\sqrt{3} - 2x = \sqrt{3} - 4 \Rightarrow x(\sqrt{3} - 2) = \sqrt{3} - 4$ **M1**

41 **DIVIDE AND RATIONALISE**

$$x = \frac{(\sqrt{3} - 4)(\sqrt{3} + 2)}{(\sqrt{3} - 2)(\sqrt{3} + 2)}$$ **M1**

42 **EXPAND**

$$= \frac{3 + 2\sqrt{3} - 4\sqrt{3} - 8}{3 - 4} = \frac{-5 - 2\sqrt{3}}{-1}$$ **M1**

ANSWER $x = 5 + 2\sqrt{3}$ **A1**

21 Histogram — the missing bar

5 marks

THE IDEA no scale? Use the bar you know: 15 people over width 10 gives frequency density 1.5 — so one big square is 1.

43 **FREQUENCIES OF THE DRAWN BARS**

$1.5 \times 10 = 15$, $4.4 \times 5 = 22$, $2 \times 15 = 30$ **M1**

44 **MISSING CLASS**

$85 - 67 = 18$ people; width $= 18 \div 0.6 = 30$ **M1**

45 **SO**

$k = 30 + 30 = 60$; draw the bar from 30 to 60 at height 0.6 **A1**

46 **(B) MORE THAN 25**

$(30 - 25) \times 2 = 10$, plus $18 = 28$ **M1**

ANSWER (a) $k = 60$ (b) $\frac{28}{85}$ **A1**

22 Rhombus — coordinate geometry

5 marks

THE IDEA the diagonals of a rhombus bisect each other at right angles, so BD is the perpendicular bisector of AC.

47 **MIDPOINT AND GRADIENT OF AC**

$M(4,6)$; $m_{AC} = \frac{8}{6} = \frac{4}{3}$, so $m_{BD} = -\frac{3}{4}$ **M1 M1**

48 **EQUATION**

$y - 6 = -\frac{3}{4}(x - 4) \Rightarrow 4y - 24 = -3x + 12 \Rightarrow 3x + 4y = 36$ **A1**

49 **B ON THE y-AXIS**

$x = 0$: $y = 9$, $B(0, 9)$. M is the midpoint of BD **M1**

ANSWER $D = (8, 3)$ **A1**

23 Three touching circles

4 marks

THE IDEA join the three centres: an equilateral triangle of side $2r$. Shaded = triangle − three 60° sectors (= half a circle).

50 **TRIANGLE**

$\frac{1}{2}(2r)(2r)\sin 60^\circ = \sqrt{3}\,r^2$ **M1**

51 **THREE SECTORS**

$3 \times \frac{60}{360}\pi r^2 = \frac{1}{2}\pi r^2$ **M1**

52 **EQUATION**

$r^2\left(\sqrt{3} - \frac{\pi}{2}\right) = 10 \Rightarrow r^2 = 62.0\ldots$ **M1**

ANSWER $r = 7.87$ **A1**

24 Probability — same colour

5 marks

THE IDEA "same colour" = RR + BB + GG. Each one is a product of two fractions.

53 **ADD THE THREE PATHS**

$$\frac{2\cdot 1+3\cdot 2+n(n-1)}{(n+5)(n+4)}=\frac{n^2-n+8}{(n+5)(n+4)}$$ **M1**

54 **SET EQUAL TO $\frac{7}{15}$**

$15(n^2-n+8)=7(n^2+9n+20)$ **M1**

55 **SIMPLIFY**

$8n^2-78n-20=0 \Rightarrow 4n^2-39n-10=0$ **A1**

56 **SOLVE**

$(4n+1)(n-10)=0$ **M1**

ANSWER $n=10$ green beads **A1**

25 3D — a line to a midpoint

5 marks

THE IDEA the point under M is halfway between the centre N and C; the height of M is half the height of V.

57 **HEIGHT OF THE PYRAMID**

$AN=\frac{1}{2}\sqrt{8^2+8^2}=4\sqrt{2}$; $VN=\sqrt{81-32}=7$ **M1**

58 **HEIGHT OF M**

3.5 **M1**

59 **HORIZONTAL DISTANCE FROM A**

$\frac{3}{4}$ of $AC=\frac{3}{4}\times 8\sqrt{2}=6\sqrt{2}=8.485\ldots$ **M1**

60 **TAN**

$$\tan\theta=\frac{3.5}{8.485\ldots}$$ **M1**

ANSWER 22.4° **A1**

26 Circle inside a sector

6 marks

THE IDEA a radius to a tangent point is perpendicular to the tangent. OP bisects the 60° angle, so there is a 30° right-angled triangle.

61 **(A) RIGHT-ANGLED TRIANGLE, ANGLE 30°**

$\sin 30^\circ = \dfrac{r}{OP} \Rightarrow OP = 2r$ **M1**

62 **(A) THE CIRCLE TOUCHES THE ARC**

radius of sector $= OP + r = 2r + r = 3r$ **A1**

63 **(B) SHADED AREA**

$\frac{60}{360}\pi(3r)^2 - \pi r^2 = \frac{3}{2}\pi r^2 - \pi r^2 = \frac{1}{2}\pi r^2$ **M1**

64 **(B) SOLVE**

$\frac{1}{2}\pi r^2 = 6\pi \Rightarrow r^2 = 12$, $r = 2\sqrt{3}$, so $R = 6\sqrt{3}$ **M1**

65 **(B) PERIMETER**

$2R + \frac{60}{360} \times 2\pi R = 12\sqrt{3} + \frac{\pi}{3}(6\sqrt{3})$ **M1**

ANSWER $12\sqrt{3} + 2\pi\sqrt{3}$ cm **A1**